Task 1: Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

a) Write a C code to create a process using fork() system call.

b) Read the user input as shell command.

c) Use a system call exec() to execute the shell command.

e) Display the PIDs of parent and child processes.

```
b.deepak@BDeepaks-MacBook-Air ~ % nano task1.c
b.deepak@BDeepaks-MacBook-Air ~ %
```

```c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <string.h>

int main() {
    char command[100];

    printf("Enter a Linux command: ");
    fgets(command, sizeof(command), stdin);

    command[strcspn(command, "\n")] = '\0';

    pid_t pid = fork();

    if (pid < 0) {
        printf("Fork failed.\n");
        return 1;
    }
    else if (pid == 0) {
        printf("Child PID: %d\n", getpid());
        execlp(command, command, NULL);
        printf("Command execution failed.\n");
        exit(1);
    }
    else {
        printf("Parent PID: %d\n", getpid());
        wait(NULL);
        printf("Child process completed.\n");
    }

    return 0;
}
```

1)Then press ctrl+oenter ctrl+x2)then gcc filename.c -o filename

```
b.deepak@BDeepaks-MacBook-Air ~ % nano task1.c
b.deepak@BDeepaks-MacBook-Air ~ % gcc task1.c -o task1
b.deepak@BDeepaks-MacBook-Air ~ %
```

3)then ./filename

```
b.deepak@BDeepaks-MacBook-Air ~ % nano task1.c
b.deepak@BDeepaks-MacBook-Air ~ % gcc task1.c -o task1
b.deepak@BDeepaks-MacBook-Air ~ % ./task1
Enter a Linux command:
```

4)enter the linux command

```
b.deepak@BDeepaks-MacBook-Air ~ % ./task1
Enter a Linux command: ls
Parent PID: 40439
Child PID: 40440
a.out                   Documents               first.c                 Music                   Pictures                task1
Applications            Downloads               Library                 os                      Public                  task1.c
Desktop                 Exp:7 DDCA.circ         Movies                  os.c                    PyCharmMiscProject      vs code
Child process completed.
b.deepak@BDeepaks-MacBook-Air ~ %
```

Child process completed.

Task 2

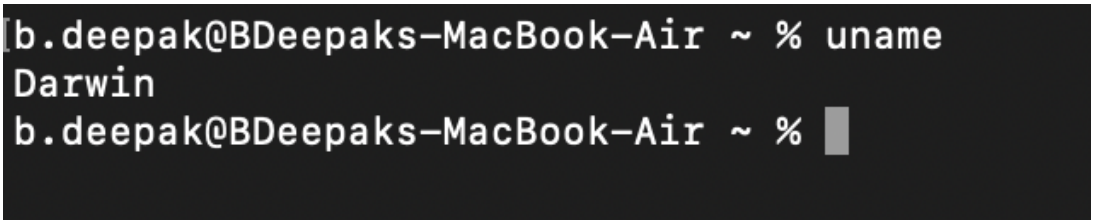
Using Linux terminal commands (uname, lscpu, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

An operating system is system software that controls all the hardware and software resources of a computer. Users do not communicate directly with the hardware. Instead, the operating system acts as a bridge between the user, applications, and the hardware

Linux Commands Used

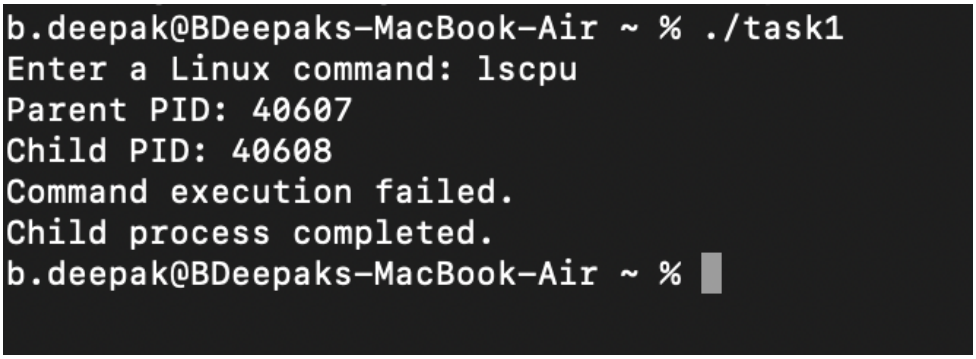
1. uname

The uname command is used to display information about the Linux operating system. It provides details such as the operating system name, kernel version, machine architecture, and hostname. This command helps us identify the system on which Linux is running.



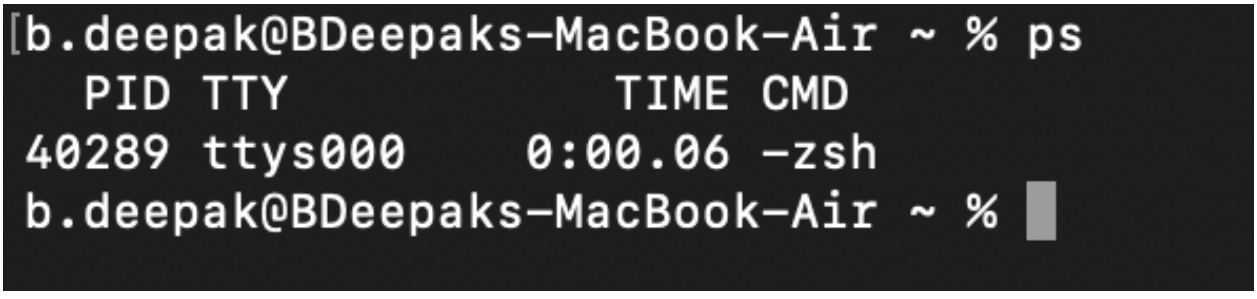
2. lscpu

The lscpu command displays detailed information about the processor. It shows the CPU model, number of cores, number of threads, CPU architecture, and processor speed. This information helps us understand the hardware capabilities of the computer.



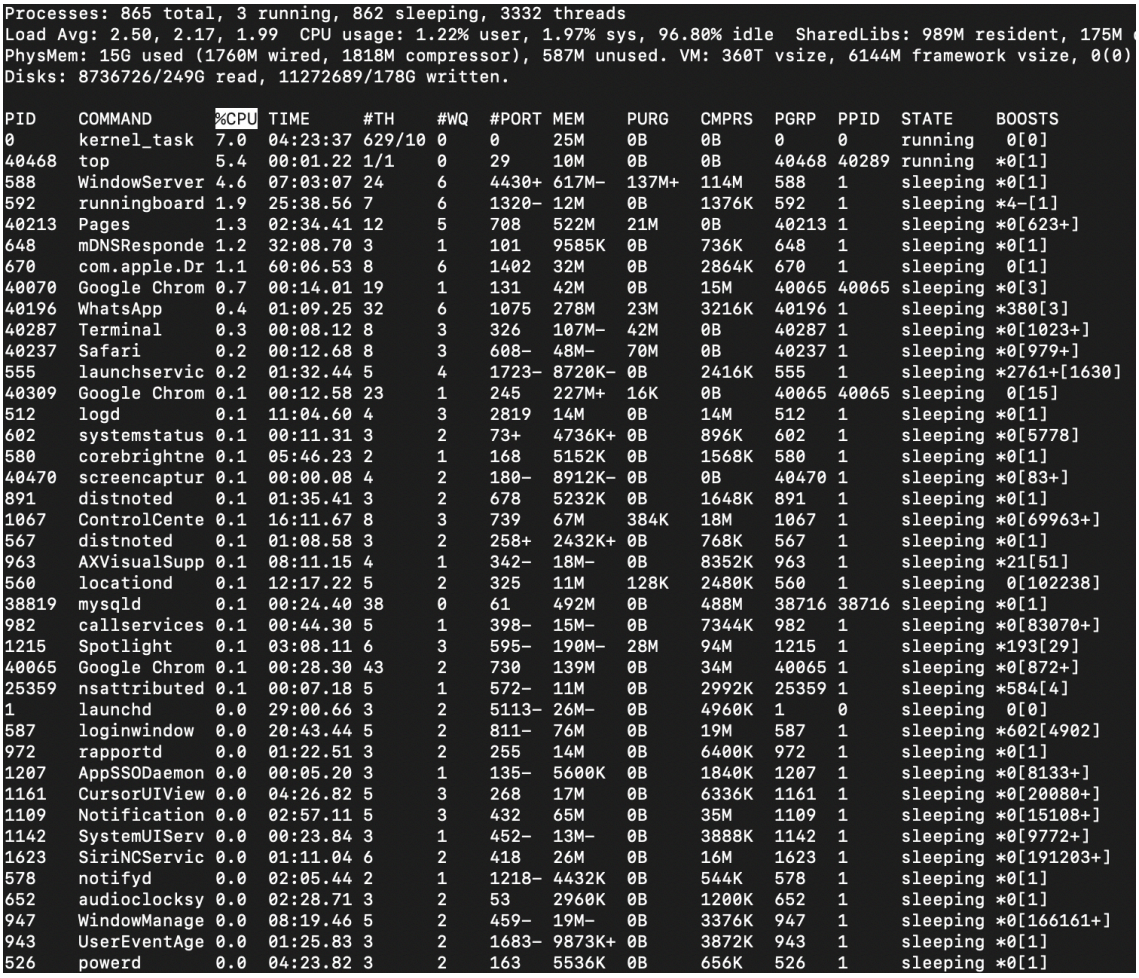
3. ps

The ps command lists the processes that are currently running in the system. It displays the Process ID (PID), Parent Process ID (PPID), user name, and the command associated with each process. This command is useful for monitoring and managing processes.



4. top

The top command provides a real-time view of the system's performance. It continuously displays CPU usage, memory usage, running processes, and system load. It is commonly used to identify programs that consume a large amount of system resources.



```
Processes: 865 total, 3 running, 862 sleeping, 3332 threads
Load Avg: 2.50, 2.17, 1.99  CPU usage: 1.22% user, 1.97% sys, 96.80% idle  SharedLibs: 989M resident, 175M
PhysMem: 15G used (1760M wired, 1818M compressor), 587M unused. VM: 360T vsize, 6144M framework vsize, 0(0)
Disks: 8736726/249G read, 11272689/178G written.

PID    COMMAND      %CPU  TIME     #TH    #WQ  #PORT MEM    PURG   CMPRS  PGRP  PPID  STATE    BOOSTS
0      kernel_task  7.0   04:23:37 629/10 0    0     25M    0B     0B     0     0     running   0[0]
40468  top          5.4   00:01.22 1/1    0    29    10M    0B     0B     40468 40289 running  *0[1]
588    WindowServer 4.6   07:03:07 24     6    4430+ 617M-  137M+  114M   588   1     sleeping *0[1]
592    runningboard 1.9   25:38.56 7      6    1320- 12M    0B     1376K  592   1     sleeping *4-[1]
40213  Pages        1.3   02:34.41 12     5    708   522M   21M    0B     40213 1     sleeping *0[623+]
648    mDNSResponde 1.2   32:08.70 3      1    101   9585K  0B     736K   648   1     sleeping *0[1]
670    com.apple.Dr 1.1   60:06.53 8      6    1402  32M    0B     2864K  670   1     sleeping  0[1]
40070  Google Chrom 0.7   00:14.01 19     1    131   42M    0B     15M    40065 40065 sleeping *0[3]
40196  WhatsApp     0.4   01:09.25 32     6    1075  278M   23M    3216K  40196 1     sleeping *380[3]
40287  Terminal     0.3   00:08.12 8      3    326   107M-  42M    0B     40287 1     sleeping *0[1023+]
40237  Safari       0.2   00:12.68 8      3    608-  48M-   70M    0B     40237 1     sleeping *0[979+]
555    launchservic 0.2   01:32.44 5      4    1723- 8720K- 0B     2416K  555   1     sleeping *2761+[1630]
40309  Google Chrom 0.1   00:12.58 23     1    245   227M+  16K    0B     40065 40065 sleeping  0[15]
512    logd         0.1   11:04.60 4      3    2819  14M    0B     14M    512   1     sleeping *0[1]
602    systemstatus 0.1   00:11.31 3      2    73+   4736K+ 0B     896K   602   1     sleeping *0[5778]
580    corebrightne 0.1   05:46.23 2      1    168   5152K  0B     1568K  580   1     sleeping *0[1]
40470  screencaptur 0.1   00:00.08 4      2    180-  8912K- 0B     0B     40470 1     sleeping *0[83+]
891    distnoted    0.1   01:35.41 3      2    678   5232K  0B     1648K  891   1     sleeping *0[1]
1067   ControlCente 0.1   16:11.67 8      3    739   67M    384K   18M    1067  1     sleeping *0[69963+]
567    distnoted    0.1   01:08.58 3      2    258+  2432K+ 0B     768K   567   1     sleeping *0[1]
963    AXVisualSupp 0.1   08:11.15 4      1    342-  18M-   0B     8352K  963   1     sleeping *21[51]
560    locationd    0.1   12:17.22 5      2    325   11M    128K   2480K  560   1     sleeping  0[102238]
38819  mysqld       0.1   00:24.40 38     0    61    492M   0B     488M   38716 38716 sleeping *0[1]
982    callservices 0.1   00:44.30 5      1    398-  15M-   0B     7344K  982   1     sleeping *0[83070+]
1215   Spotlight    0.1   03:08.11 6      3    595-  190M-  28M    94M    1215  1     sleeping *193[29]
40065  Google Chrom 0.1   00:28.30 43     2    730   139M   0B     34M    40065 1     sleeping *0[872+]
25359  nsattributed 0.1   00:07.18 5      1    572-  11M    0B     2992K  25359 1     sleeping *584[4]
1      launchd      0.0   29:00.66 3      2    5113- 26M-   0B     4960K  1     0     sleeping  0[0]
587    loginwindow  0.0   20:43.44 5      2    811-  76M    0B     19M    587   1     sleeping *602[4902]
972    rapportd     0.0   01:22.51 3      2    255   14M    0B     6400K  972   1     sleeping *0[1]
1207   AppSSODaemon 0.0   00:05.20 3      1    135-  5600K  0B     1840K  1207  1     sleeping *0[8133+]
1161   CursorUIView 0.0   04:26.82 5      3    268   17M    0B     6336K  1161  1     sleeping *0[20080+]
1109   Notification 0.0   02:57.11 5      3    432   65M    0B     35M    1109  1     sleeping *0[15108+]
1142   SystemUIServ 0.0   00:23.84 3      1    452-  13M-   0B     3888K  1142  1     sleeping *0[9772+]
1623   SiriNCServic 0.0   01:11.04 6      2    418   26M    0B     16M    1623  1     sleeping *0[191203+]
578    notifyd      0.0   02:05.44 2      1    1218- 4432K  0B     544K   578   1     sleeping *0[1]
652    audioclocksy 0.0   02:28.71 3      2    53    2960K  0B     1200K  652   1     sleeping *0[1]
947    WindowManage 0.0   08:19.46 5      2    459-  19M-   0B     3376K  947   1     sleeping *0[166161+]
943    UserEventAge 0.0   01:25.83 3      2    1683- 9873K+ 0B     3872K  943   1     sleeping *0[1]
526    powerd       0.0   04:23.82 3      2    163   5536K  0B     656K   526   1     sleeping *0[1]
```